

Fig 7.7

$P(1.25 < Z < 2.75) = P(0 < Z < 2.75) - P(0 < Z < 1.25)$

$= 0.4970 - 0.3944 = 0.1026$

Example 7.22

If X is a normal variate with mean 30 and SD 5. Find the probabilities that (i) $26 \leq X \leq 40$ (ii) $X > 45$

Solution :

Here mean μ= 30 and
standard deviation σ = 5

(i) When $X = 26$ $\quad Z = (X - \mu)/\sigma$

$= (26 - 30)/5 = -0.8$

And when $X = 40$, $\quad Z = \dfrac{40-30}{5} = 2$

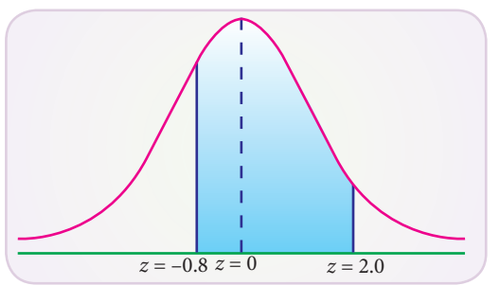


Fig 7.8

Therefore,

$P(26<X<40)) = P(-0.8 \leq Z < 2)$

$= P(-0.8 \leq Z \leq 0) + p(0 \leq Z \leq 2)$

$= P(0 \leq Z \leq 0.8) + P(0 \leq Z \leq 2)$

$= 0.2881 + 0.4772$ (By tables)

$= 0.7653$

(ii) The probability that $X \geq 45$

When $X = 45$

$$Z = \frac{X-\mu}{\sigma} = \frac{45-30}{5} = 3$$

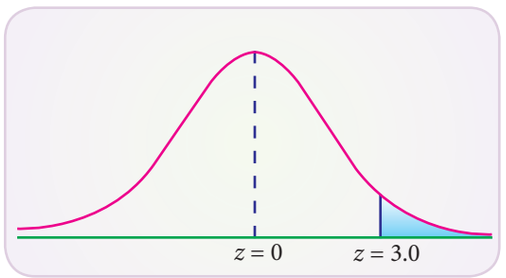


Fig 7.9

$P(X \geq 45) = P(Z \geq 3)$

$= 0.5 - 0.49865$

$= 0.00135$

Example 7.23

The average daily sale of 550 branch offices was ₹ 150 thousand and standard deviation is ₹ 15 thousand. Assuming the distribution to be normal, indicate how many branches have sales between

(i) ₹ 1,25,000 and ₹ 1, 45, 000

(ii) ₹ 1,40,000 and ₹ 1,60,000

Solution :

Given that mean μ= 150 and standard deviation σ= 15

(i) when $X = 125$ thousand

$$Z = \frac{X-\mu}{\sigma} = \frac{125-150}{15} = -1.667$$

When $X = 145$ thousand

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